

Math vs. Your Eyes

 **Green Level: I Want More Support**

Name: _____ Date: _____ Period: _____

Can Math Prove Your Eyes Wrong?

Your eyes can be tricked by optical illusions—but a ruler never lies! Today you'll use math to prove that what you SEE isn't always what's REAL.

Activity 1: The Müller-Lyer Illusion

Look at these two lines. Which one looks LONGER?



My prediction: (circle one)

Top line is longer | Bottom line is longer | Same length

Now measure with your ruler!

Top line: _____ cm Bottom line: _____ cm

Was your prediction correct? Yes / No

Why does this illusion work? The arrows create *angles* that trick your brain into thinking one line is closer (and shorter) than the other!

Activity 2: Simple Perspective Drawing

Follow these steps to draw a 3D box using math:

1. Put a dot in the center of the box below (this is your **vanishing point**)
2. Draw a square in the bottom-left corner
3. Draw lines from each corner of the square to the vanishing point
4. Draw a smaller square where those lines would end

You just created the illusion of depth using geometry!

Draw your perspective box here

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What Did You Learn?

Fill in the blanks:

The two lines in the Müller-Lyer illusion are actually _____ (the same / different).

The arrows create _____ that trick your brain.

When I used a ruler, I found out that math is more _____ than my eyes.

Why Does Math Matter?

Think of times when measuring is more reliable than just looking:



Mini Museum Connection

The stereoscope in the Mini Museum uses math too! Your eyes are about 6 cm apart, so each eye sees a slightly different angle. The stereoscope shows each eye a different picture, and your brain calculates depth from the difference!

Visit: minimuseumproject.com/exhibitions/seeing-is-deceiving

Try This Math Problem!

The stereoscope cards were photographed with cameras 6.5 cm apart (the average distance between human eyes).

If YOUR eyes are 6 cm apart, what is the difference? _____ cm

What percentage is that? (Hint: divide the difference by 6.5, then multiply by 100)

Answer: _____ %

Share with Your Family!

Show your family the Müller-Lyer illusion. Ask them which line is longer, then measure to prove the truth! See whose eyes are tricked the most.